

MATH 3260 - Chapter 3 Notes: Determinants

Feedback-Revised Student Notes

Kennesaw State University - MATH 3260

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These notes are synchronized with the feedback-revised student notebook. They emphasize definitions, theorems, worked mathematics, conceptual connections, and interpretation while omitting notebook-specific code and worksheet instructions.

Chapter overview

1. **Section 3.1:** The Determinant of a Matrix
2. **Section 3.2:** Determinants and Elementary Operations
3. **Section 3.3:** Properties of Determinants
4. **Section 3.4:** Applications of Determinants

Each section ends with a terminology table, a worksheet, and complete instructor solutions.

Example policy

All worked examples and numerical values are original. The chapter uses personal-interest themes selectively, including the Spurs, hiking, travel, badminton, driving, Backstreet Boys, and R&B music. Any prices, route coordinates, or datasets are fictional teaching models unless explicitly identified otherwise.

Section 3.1 - The Determinant of a Matrix

Learning objectives

By the end of this section, students should be able to:

- compute the determinant of a 2×2 matrix;
- interpret a determinant as signed area scaling;
- find minors and cofactors;
- use cofactor expansion along any row or column;
- select an efficient expansion row or column;
- evaluate determinants of triangular matrices.

Definition: determinant of a 2 by 2 matrix

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the determinant is $\det(A) = ad - bc$. The notation $|A|$ is also used, but it should not be confused with absolute value.

Original example: three 2×2 determinants

Consider

$$A = \begin{bmatrix} 9 & 1 \\ 20 & 21 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 6 \\ 2 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} -2 & 5 \\ 7 & 1 \end{bmatrix}.$$

For A ,

$$\det(A) = 9(21) - 1(20) = 189 - 20 = 169.$$

For B ,

$$\det(B) = 4(3) - 6(2) = 12 - 12 = 0.$$

For C ,

$$\det(C) = (-2)(1) - 5(7) = -2 - 35 = -37.$$

A zero determinant signals that the corresponding linear transformation collapses area and is not invertible.

Geometric meaning: signed area scaling

The columns of a 2×2 matrix determine a parallelogram. Its area is

$$|\det(A)|.$$

The sign records orientation:

- $\det(A) > 0$: orientation is preserved;
- $\det(A) < 0$: orientation is reversed;
- $\det(A) = 0$: the parallelogram collapses to a line or point.

Definition: minor and cofactor

For a square matrix A , the minor M_{ij} is the determinant obtained after deleting row i and column j . The cofactor is $C_{ij} = (-1)^{i+j}M_{ij}$. The cofactor signs alternate in a checker-board pattern.

Why minors and cofactors are introduced

A 3×3 determinant is reduced to several 2×2 determinants. A **minor** is obtained by deleting one row and one column; a **cofactor** adds the alternating sign pattern. These ideas reappear later in the adjugate formula and Cramer's Rule.

The 3×3 cofactor sign pattern is

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}.$$

Original example: all minors and cofactors

Let

$$H = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix}.$$

To find M_{12} , delete row 1 and column 2:

$$M_{12} = \det \begin{bmatrix} -1 & 2 \\ 4 & 1 \end{bmatrix} = (-1)(1) - 2(4) = -9.$$

Because $(-1)^{1+2} = -1$,

$$C_{12} = -M_{12} = 9.$$

The complete minor matrix is

$$[M_{ij}] = \begin{bmatrix} 3 & -9 & -12 \\ 1 & 2 & -4 \\ 2 & 4 & 7 \end{bmatrix},$$

and the cofactor matrix is

$$[C_{ij}] = \begin{bmatrix} 3 & 9 & -12 \\ -1 & 2 & 4 \\ 2 & -4 & 7 \end{bmatrix}.$$

Cofactor expansion

For an $n \times n$ matrix, expansion along row i is

$$\det(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}.$$

Expansion along column j is

$$\det(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}.$$

The determinant is the same regardless of the selected row or column. Choose a row or column containing many zeros.

Detailed expansion of H along the first row

Because the third entry in the first row is zero,

$$\det(H) = 2C_{11} + 1C_{12} + 0C_{13}.$$

We already found $C_{11} = 3$ and $C_{12} = 9$, so

$$\det(H) = 2(3) + 1(9) + 0(-12) = 15.$$

The key pattern is

$$\det(H) = h_{11} \det(M_{11}) - h_{12} \det(M_{12}) + h_{13} \det(M_{13}),$$

and every 2×2 minor then uses $ad - bc$. A common 3×3 error is forgetting the negative sign on the middle cofactor.

Original 4×4 example: exploit a sparse column

Let

$$Q = \begin{bmatrix} 2 & 1 & 0 & 3 \\ 0 & -1 & 0 & 4 \\ 5 & 2 & 7 & 1 \\ 1 & 0 & 0 & 2 \end{bmatrix}.$$

The third column has only one nonzero entry, $q_{33} = 7$. Therefore,

$$\det(Q) = 7C_{33}.$$

Since $(-1)^{3+3} = 1$,

$$C_{33} = \det \begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 1 & 0 & 2 \end{bmatrix}.$$

Expand this 3×3 determinant along the second row. The middle entry is -1 :

$$(-1)(+1) \det \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} = (-1)(4 - 3) = -1.$$

The last entry is 4 and its cofactor sign is negative:

$$4(-1) \det \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} = 4(-1)(-1) = 4.$$

Therefore,

$$C_{33} = -1 + 4 = 3.$$

Hence

$$\det(Q) = 7(3) = 21.$$

Theorem: determinant of a triangular matrix

If A is upper triangular or lower triangular, then $\det(A)$ is the product of the diagonal entries.

For

$$T = \begin{bmatrix} 3 & 0 & 0 & 0 \\ -2 & 5 & 0 & 0 \\ 1 & 4 & -2 & 0 \\ 7 & 1 & 6 & 4 \end{bmatrix},$$

we have

$$\det(T) = 3(5)(-2)(4) = -120.$$

Common mistakes

Do not apply the $ad - bc$ rule to matrices larger than 2×2 . Do not forget the alternating cofactor signs. During cofactor expansion, multiply each entry by its cofactor, not merely by its minor.

Definitions and terminology

Term	Meaning
Determinant	Scalar associated with a square matrix
Signed area scale	Geometric interpretation of a 2×2 determinant
Minor M_{ij}	Determinant after deleting row i and column j
Cofactor C_{ij}	$(-1)^{i+j} M_{ij}$
Cofactor expansion	Sum of entries times cofactors along a row or column
Upper triangular	Zeros below the main diagonal
Lower triangular	Zeros above the main diagonal
Diagonal matrix	Zeros above and below the main diagonal

Section 3.2 - Determinants and Elementary Operations

Learning objectives

Students should be able to:

- track determinant changes under row operations;
- track determinant changes under column operations;
- evaluate determinants efficiently by reducing to triangular form;
- recognize structural conditions that force a zero determinant;
- connect a zero determinant with redundant information.

Theorem: elementary row operations

Interchanging two rows changes the sign of the determinant. Adding a multiple of one row to another leaves the determinant unchanged. Multiplying one row by c multiplies the determinant by c . The same rules hold for column operations.

Detailed example: triangularization without changing the determinant

Let

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 4 & 3 & 0 \\ -2 & 2 & 5 \end{bmatrix}.$$

Start with

$$\det(A) = \det \begin{bmatrix} 2 & 1 & -1 \\ 4 & 3 & 0 \\ -2 & 2 & 5 \end{bmatrix}.$$

Apply

$$R_2 \leftarrow R_2 - 2R_1.$$

Adding a multiple of one row to another does not change the determinant:

$$\det(A) = \det \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 2 \\ -2 & 2 & 5 \end{bmatrix}.$$

Next apply

$$R_3 \leftarrow R_3 + R_1,$$

giving

$$\det(A) = \det \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 2 \\ 0 & 3 & 4 \end{bmatrix}.$$

Finally apply

$$R_3 \leftarrow R_3 - 3R_2,$$

giving

$$\det(A) = \det \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & -2 \end{bmatrix}.$$

The matrix is upper triangular, so

$$\det(A) = 2(1)(-2) = -4.$$

Row interchange and row scaling

If B is obtained by swapping two rows of A , then

$$\det(B) = -\det(A).$$

If C is obtained by multiplying one row of A by 5, then

$$\det(C) = 5 \det(A).$$

If several row operations are used, keep a determinant accounting record. A common strategy is to avoid scaling rows until the final triangular matrix is reached.

Determinant effect of the row operations

$$\begin{array}{l|l} R_i \leftarrow R_i + cR_j & \det(B) = \det(A) \\ R_i \leftrightarrow R_j & \det(B) = -\det(A) \\ R_i \leftarrow cR_i & \det(B) = c \det(A) \end{array}$$

Column operations and zero determinants

Consider

$$D = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 8 & 5 \\ 2 & 4 & 7 \end{bmatrix}.$$

The second column is twice the first column. Apply

$$C_2 \leftarrow C_2 - 2C_1.$$

This column replacement does not change the determinant, and it creates a zero column:

$$\det(D) = \det \begin{bmatrix} 1 & 0 & 3 \\ 4 & 0 & 5 \\ 2 & 0 & 7 \end{bmatrix} = 0.$$

A zero determinant is also a dependence test: if rows or columns are linearly dependent, then $\det(A) = 0$.

Conditions that force a zero determinant

A square matrix has determinant zero if it has an all-zero row or column, two equal rows or columns, or one row or column that is a scalar multiple of another. More generally, any linear dependence among the rows or columns forces determinant zero.

Application: redundant data features

Suppose columns of a data matrix represent three recorded features, but the second feature is exactly twice the first. Then the columns are linearly dependent. The determinant is zero, so the matrix cannot be inverted. In modeling terms, the second feature contributes no new independent information.

Definitions and terminology

Term	Meaning
Row interchange	Swap two rows; determinant changes sign
Row replacement	Add a multiple of one row to another; determinant unchanged
Row scaling	Multiply one row by c ; determinant multiplied by c
Column-equivalent	Related by elementary column operations
Zero determinant	Determinant equal to zero; matrix is singular
Redundant information	A row or column depends on others

Section 3.3 - Properties of Determinants

Learning objectives

Students should be able to:

- compute determinants of matrix products without multiplying first;
- use the scalar-multiple rule $\det(cA) = c^n \det(A)$;
- compute determinants of inverses and powers;
- connect nonzero determinant with invertibility;
- use $\det(A^T) = \det(A)$;
- interpret determinant rules through composition of transformations.

Theorem: determinant of a product

For square matrices of the same order, $\det(AB) = \det(A) \det(B)$. Matrix multiplication is generally not commutative, but determinants satisfy $\det(AB) = \det(BA)$.

Original example: product property

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 \\ 4 & 2 \end{bmatrix}.$$

Then

$$\det(A) = 2(3) - 1(1) = 5,$$

and

$$\det(B) = 1(2) - (-1)(4) = 6.$$

Therefore,

$$\det(AB) = \det(A) \det(B) = 5(6) = 30.$$

Indeed,

$$AB = \begin{bmatrix} 6 & 0 \\ 13 & 5 \end{bmatrix},$$

so

$$\det(AB) = 6(5) - 0(13) = 30.$$

Why $\det(AB) = \det(BA)$

$$\det(AB) = \det(A) \det(B) = \det(B) \det(A) = \det(BA).$$

This does **not** imply $AB = BA$.

Geometric interpretation of the product rule

If B scales area by $\det(B)$ and then A scales the resulting area by $\det(A)$, the combined transformation AB scales area by the product

$$\det(A) \det(B).$$

This is why the determinant of a composition is multiplicative.

Theorem: scalar multiples

If A is an $n \times n$ matrix, then $\det(cA) = c^n \det(A)$. Multiplying the entire matrix by c multiplies each of its n rows by c .

For the upper triangular matrix

$$C = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix},$$

we have

$$\det(C) = 1(3)(2) = 6.$$

Because C is 3×3 ,

$$\det(2C) = 2^3 \det(C) = 8(6) = 48.$$

Invertibility and inverse determinants

A square matrix is invertible if and only if its determinant is nonzero. If A is invertible, then $\det(A^{-1}) = 1/\det(A)$.

For

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix},$$

we found $\det(A) = 5$. Therefore,

$$\det(A^{-1}) = \frac{1}{5}.$$

This follows from

$$1 = \det(I) = \det(AA^{-1}) = \det(A) \det(A^{-1}).$$

A nontriangular scalar-multiple example

For

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}, \quad \det(A) = -1,$$

we have

$$\det(4A) = -16 = 4^2 \det(A),$$

illustrating $\det(cA) = c^n \det(A)$ beyond the triangular case.

Equivalent conditions for a nonsingular matrix

For an $n \times n$ matrix A , the following conditions are equivalent:

1. A is invertible;
2. $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b}$;
3. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution;
4. A is row-equivalent to I_n ;
5. A is a product of elementary matrices;
6. $\det(A) \neq 0$.

A single determinant calculation can therefore answer several structural questions.

These conditions are interchangeable tests for the same structure. For example, a pivot in every column implies invertibility, hence $\det(A) \neq 0$, independent columns, and only the trivial solution to $A\mathbf{x} = \mathbf{0}$.

Theorem: transpose

For every square matrix, $\det(A^T) = \det(A)$. Transposition exchanges rows and columns but preserves the determinant.

Let

$$R = \begin{bmatrix} 2 & 0 & 1 \\ -1 & 3 & 2 \\ 4 & 1 & 0 \end{bmatrix}.$$

A direct calculation gives

$$\det(R) = -13.$$

Its transpose is

$$R^T = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 1 \\ 1 & 2 & 0 \end{bmatrix},$$

and

$$\det(R^T) = -13.$$

Determinants of powers

Repeated use of the product theorem gives

$$\det(A^k) = \det(A)^k$$

for every positive integer k . For invertible A , the rule also extends to negative integer powers.
 For the matrix A above with $\det(A) = 5$,

$$\det(A^4) = 5^4 = 625.$$

Definitions and terminology

Term	Meaning
Product rule	$\det(AB) = \det(A) \det(B)$
Scalar-multiple rule	$\det(cA) = c^n \det(A)$ for $n \times n$ A
Nonsingular	Invertible; determinant nonzero
Singular	Not invertible; determinant zero
Inverse determinant	$\det(A^{-1}) = 1/\det(A)$
Transpose determinant	$\det(A^T) = \det(A)$
Determinant of a power	$\det(A^k) = \det(A)^k$

Section 3.4 - Applications of Determinants

Learning objectives

Students should be able to:

- form the cofactor matrix and adjugate;
- find an inverse using the adjugate formula;
- solve square systems using Cramer's Rule;
- compute triangle area and test collinearity;
- write a determinant equation for a line;
- compute tetrahedron volume and test coplanarity;
- write an equation of a plane through three points.

Definition: adjugate

The adjugate of A , denoted $\text{adj}(A)$, is the transpose of the cofactor matrix. If $\det(A) \neq 0$, then $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$.

Detailed adjugate example

Let

$$M = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

The cofactor matrix is

$$C = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}.$$

Therefore,

$$\text{adj}(M) = C^T = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}.$$

Since $\det(M) = 2$,

$$M^{-1} = \frac{1}{2} \text{adj}(M) = \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}.$$

The adjugate is useful because

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A), \quad \det(A) \neq 0.$$

So the cofactor calculations from Section 3.1 directly build an inverse.

Cramer's Rule

For $A\mathbf{x} = \mathbf{b}$ with $\det(A) \neq 0$, replace column i of A with $\mathbf{b}$ to form A_i . Then $x_i = \det(A_i)/\det(A)$.

Original application: concert merchandise prices

At a fictional Backstreet Boys fan event, let

- p = poster price;
- w = wristband price;
- t = tour-book price.

Three bundle totals give

$$\begin{cases} p + w = 29, \\ w + t = 41, \\ p + t = 30. \end{cases}$$

The coefficient matrix and constant vector are

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 29 \\ 41 \\ 30 \end{bmatrix}.$$

We have

$$D = \det(A) = 2.$$

Replace the first column by $\mathbf{b}$:

$$D_p = \det \begin{bmatrix} 29 & 1 & 0 \\ 41 & 1 & 1 \\ 30 & 0 & 1 \end{bmatrix} = 18.$$

Similarly,

$$D_w = 40, \quad D_t = 42.$$

Therefore,

$$p = \frac{18}{2} = 9, \quad w = \frac{40}{2} = 20, \quad t = \frac{42}{2} = 21.$$

Area of a triangle and collinearity

For vertices (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) ,

$$\text{Area} = \frac{1}{2} \left| \det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} \right|.$$

Hiking-map example

Three trail landmarks have coordinates

$$(1, 2), \quad (7, 5), \quad (4, 10).$$

Then

$$\det \begin{bmatrix} 1 & 2 & 1 \\ 7 & 5 & 1 \\ 4 & 10 & 1 \end{bmatrix} = 39,$$

so

$$\text{Area} = \frac{|39|}{2} = \frac{39}{2} = 19.5$$

square map units. Three points are collinear exactly when this determinant is zero.

If the determinant-based area is zero, the triangle has zero area, so the three points are collinear.

Determinant equation of a line

A point (x, y) lies on the line through (x_1, y_1) and (x_2, y_2) when

$$\det \begin{bmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{bmatrix} = 0.$$

For travel-map points $(2, 5)$ and $(8, 17)$,

$$\det \begin{bmatrix} x & y & 1 \\ 2 & 5 & 1 \\ 8 & 17 & 1 \end{bmatrix} = 0.$$

Expanding gives

$$-12x + 6y - 6 = 0,$$

or

$$y = 2x + 1.$$

Volume of a tetrahedron

For four points in space, the tetrahedron volume is

$$V = \frac{1}{6} \left| \det \begin{bmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{bmatrix} \right|.$$

For

$$(0, 0, 0), \quad (2, 0, 0), \quad (0, 3, 0), \quad (0, 0, 5),$$

we obtain

$$V = \frac{1}{6}|30| = 5$$

cubic units.

Equation of a plane through three points

A point (x, y, z) lies in the plane through three noncollinear points when

$$\det \begin{bmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{bmatrix} = 0.$$

Use the points

$$P_1 = (1, 0, 2), \quad P_2 = (3, 1, 0), \quad P_3 = (0, 2, 1).$$

Vectors in the plane are

$$P_2 - P_1 = (2, 1, -2), \quad P_3 - P_1 = (-1, 2, -1).$$

Their cross product is

$$(2, 1, -2) \times (-1, 2, -1) = (3, 4, 5).$$

Therefore, the plane has equation

$$3(x - 1) + 4(y - 0) + 5(z - 2) = 0,$$

or

$$3x + 4y + 5z = 13.$$

A fourth point is coplanar with these three exactly when it satisfies this equation, equivalently when the corresponding 4×4 determinant is zero.

In the example, $(3, 4, 5)$ is a **normal vector** to the plane.

Definitions and terminology

Term	Meaning
Cofactor matrix	Matrix $[C_{ij}]$
Adjugate	Transpose of the cofactor matrix
Adjugate inverse formula	$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$
Cramer's Rule	$x_i = \det(A_i) / \det(A)$

Term	Meaning
Collinear	Points lying on one line
Coplanar	Points lying in one plane
Determinant line equation	3×3 determinant equal to zero
Determinant plane equation	4×4 determinant equal to zero
Tetrahedron volume	One sixth of an absolute 4×4 determinant

Chapter 3 Summary

The chapter develops the determinant from a computational formula into a structural and geometric tool:

minor and cofactor calculations $\rightarrow$ efficient row-operation evaluation $\rightarrow$ invertibility and product properties

Core habits:

1. use $ad - bc$ only for 2×2 matrices;
2. choose sparse rows or columns for cofactor expansion;
3. track every row or column operation affecting the determinant;
4. remember the exponent in $\det(cA) = c^n \det(A)$;
5. connect $\det(A) = 0$ with singularity and dependence;
6. use absolute values for geometric area and volume;
7. use Python to verify calculations, not replace written reasoning.

Clarification: transpose is not a row interchange

The identity

$$\det(A^T) = \det(A)$$

does not contradict the rule that swapping two rows changes the sign of a determinant. A row interchange changes only two rows while leaving the remaining row structure fixed. Transposition performs a different operation: it exchanges the entire row structure with the column structure. The determinant is symmetric with respect to rows and columns, so its value is preserved under transposition.